The course and timings-

Course Number and Title	**CHL4380 - Applications of Artificial Intelligence in Chemical Engineering**	
Lectures	Days	Tuesday, Thursday and Friday
	Time	4:00 PM to 4:50 PM
	Venue	MT 110

Instructor(s)-

Name	**Email**	**Department**	**Contact Number**	**Office Address**
Course Coordinator				
Varanasi Santhosh Kumar	**skvaranasi@iitj.ac.in**	**Chemical Engineering**	**02912801719**	**325, Second floor, Chemistry building.**

Office Timings during the week: Friday 11:00 AM to 12:00 PM

Contact details of Teaching Assistants (TAs): Please contact the TAs for any query related to the course or attendance.

Name of the TA	Contact details
Mr. Nisargendu H. Bhatt	p25ch0011@iitj.ac.in

Evaluation policy for the course:

Components	**Weightage (%)**	**Date and Timings**	**Remarks**
Minor	15%	TBD	Written Examination
Major	40%	TBD	Closed Book, Written examination, Scientific calculators are permitted
Quizzes and Assignments	**20%**	**TBD**	
Course Project	**25%**	**TBD**	

Attendance Policy:

75% attendance complying with Institute's attendance policy. Attendance will be taken at the beginning of every class and is made available to the students via google classroom.

Lecture-wise detailed course content as approved by the Senate:

Title	**Applications of Artificial Intelligence in Chemical Engineering**	Number	CHL4380
Department	Chemical Engineering	L-T-P [C]	3 -0 -0
Offered for	B.Tech, Dual Degree	Type	Programme Elective
Prerequisite	Material and Energy Balance, Scientific Computations		

Objectives

The instructor will:

1. Provide background on AI in chemical engineering and allied systems

Learning Outcomes

The students are expected to:

1. Gain fundamental understanding of the application of AI in chemical and allied engineering.
2. Learn to develop AI model equations, approaches for chemical and allied engineering systems.
3. Learn to write basic codes of AI for simple systems

Course Content

Artificial Intelligence in Chemical Engineering Background [5 lectures]: *Early Attempts, Expert System Era, Neural Network Era, Effects and Lacunae of Expert System Era and Neural Network Era on Chemical Engineering, Deep Learning and Data Science Era.*

AI in modelling and Optimization [20 lectures]*: AI in chemical process modelling, Modelling Real-World Processes: Deep and Shallow Knowledge Integrated with Approximate Reasoning in a Diagnostic Expert System, AI in optimization of chemical process, Case studies: Oxidative coupling of methane (OCM) over the Mn/Na2WO4/SiO2 catalyst in a fluidized bed reactor, Design of Protein Purification Processes by Heuristic Search.*

AI in Control [5 lectures]: *Application of neural networks in chemical process control, Control of Reactive Distillation Column.*

AI techniques in fault detection and diagnosis of chemical engineering [8 lectures]*: Fault Detection and Diagnosis Using Artificial Neural Networks, A Modular Approach to Multiple Faults Diagnosis, Qualitative Models in Discrete Event Simulation to Analyse Malfunctions in Processing Systems.*

AI *in Chemical Engineering Recent Trends and Future Outlook* : [1 lecture]*: Emergence in large scale systems of self-organizing intelligent agents.*

Text Books:

1. Michael L. Mavrovouniotis, 1990, Artificial Intelligence in Process Engineering, Academic Press

Reference Books:

1. Venkat Venkatasubramanian, 2019, The Promise of Artificial Intelligence in Chemical Engineering: Is It Here, finally? AIChE, Vol. 65, No. 2
2. Zeinab Hajjar, Shokoufe Tayyebi and Mohammad Hosein Eghbal Ahmadi, 2018, Application of AI in Chemical Engineering.
3. Eghbal-Ahmadi M-H, Zaerpour M, Daneshpayeh M, Mostoui N. Optimization of fluidized bed reactor of oxidative coupling of methane. International Journal of Chemical Reactor Engineering. 2012; 10:1-21

4. Araromi DO, Sonibare JA, Emuoyibofarhe JO. Fuzzy identification of reactive distillation for acetic acid recovery from waste water. Journal of Environmental Chemical Engineering. 2014; 2:1394-1403
5. Shih-Bo Hung and Ming-Jer Lee, 2006, Control of Different Reactive Distillation Configurations, AIChE, Vol. 52, No. 4
6. Neha Sharma and Kailash Singh, 2010, Control of Reactive Distillation Column: A Review, International Journal of Chemical Reactor Engineering, Vol. 8